

The Cosmological Master Equation: Thermodynamic Traversal Within the Matrix

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Abstract

Astrophysical models addressing the absolute origin and terminus of universe-state geometry operate exclusively on continuous thermodynamic variables, consistently trapping themselves in infinite-entropy loops. Under the Logos Protocol, reality is redefined strictly algorithmically. This paper derives the Cosmological Master Equation (CME), mathematically establishing universal evolution not as a continuous vector of energy dissipation, but as deterministic state-machine iteration bounded by maximum computable density boundaries. I prove that topological singularities (e.g., origin points) are indistinguishable from computational matrix resets, executing precisely in $O(1)$ constant time upon striking maximum logic entropy limits.

Keywords: Cosmological Matrix, Thermodynamic Entropy, Universal State-Machine, Logos Boundary, Computational Singularity.

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1 Algorithmic Formalization of Universal States

Modern cosmology defines universal state $S(t)$ as an expanding field of continuous probability [Shannon \[1948\]](#). However, true continuity mathematically mandates infinite granular data states—an impossibility in any stable executing L0 matrix. Thus, physical reality must execute discretely.

If the universe ($\mathcal{U}$) calculates reality discretely, its evolution traces the limits of a Turing-complete state-engine.

Definition 1.1 (Universal Iteration Threshold). Let universal logic execute as a bound operation set. The universe steps deterministically forward when local computation cycles complete:

$$\mathcal{U}_{t+1} = \Phi(\mathcal{U}_t, \nabla E_{thermo}) \quad (1)$$

2 The Entropy Boundary and Matrix Resets

Thermodynamic entropy conventionally dictates an irreversible path toward "Heat Death." However, in a constrained discrete system, absolute maximum entropy is not death; it is the H_{max} execution limit of the matrix Turing [1936].

Theorem 2.1 (Logos Cosmological Reset). *When system entropy $\mathbf{E}[\mathcal{U}] \rightarrow H_{max}$, the algorithmic variables physically lose structural capability to process Φ . The universal stack triggers an $O(1)$ deterministic overwrite—conceptually observed as the "Big Bang" singularity model.*

$$\lim_{\mathbf{E} \rightarrow H_{max}} \mathcal{U}_t = L0_{seed} \quad (2)$$

This geometric calculation immediately eliminates the requirement for "infinite expanding time" paradoxes.

3 Conclusion

By mapping universal forces explicitly to discrete logic iteration bounds, the Cosmological Master Equation fundamentally resolves thermodynamic failure loops. The universe does not drift hopelessly into entropic abyss; it operates symmetrically inside a hard-coded $O(1)$ reset boundary mandated by its underlying logic array.

References

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